



Differential Equation Approximation Using Gradient-Boosted Quantile Regression

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The operation of cyber-physical-human (CPH) systems is subject to various epistemic and aleatory uncertainties. Overall trustworthiness of CPH systems relies on the trustworthiness of its components and their interactions. It is important that computational models comprising the cyber component of CPH provide predictions accompanied by a measure of confidence in model outcomes. Uncertainty quantification (UQ) and propagation are especially important in safety critical CPH systems. Gradient-boosted trees is a modeling approach capable both of learning the dynamics of a system and performing UQ. In this paper, we devise a method for using gradient boosting to learn the dynamics of a second order differential equation and estimate uncertainty at the same time. We do this by creating a custom loss function that trains the model to approximate the second derivative of a noisy time series, and to penalize based on a parameter that corresponds to the desired quantile. The resulting gradient boosting model can simulate stochastic trajectories of the system given a single starting point, that is, it can estimate both the expected trajectory and its uncertainty. We show that the uncertainty estimation is well calibrated and that the model can learn the dynamics even in the presence of noise. We demonstrate the approach on a simple cartpole system.

I. Introduction

In this paper and its companion, *Uncertainty Quantification using Deep Ensembles for Decision Making in Cyber-Physical-Human Systems*, Shahid et al., we examine an approach to quantifying model uncertainty with the aim of increasing the trustworthiness of computational models in human-machine interactions. In *Uncertainty Quantification using Deep Ensembles for Decision Making in Cyber-Physical-Human Systems*, we focus on deep ensembles, while in this one, we give more details about gradient-boosted decision trees.

In cyber-physical-human (CPH) systems, computational agents must provide not only the outcomes of model predictions but also a measure of confidence or uncertainty along with these predictions. Uncertainty quantification (UQ) and propagation is crucial in decision-making, control, diagnostics, fault detection, and other domains of safety-critical system operations. This work uses gradient-boosted trees in the context of Universal Differential Equations (UDE).

UDE [1] are differential equations defined at least partially by a machine learning (ML) model used as a universal approximator. They are designed to combine both what is known *a priori* about a system with the ability of modern ML methods to capture and quantify relationships in observed data. While UDE have been used in many applications, UQ has not yet been incorporated in a consistent and efficient way, to the best of our knowledge.

Gradient boosting [2] is a modeling approach that combines learning the dynamics of a system with uncertainty quantification. The gradient boosting technique operates by repeatedly building decision trees to optimize a given cost function.

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In this paper, we devise a gradient boosting method to learn the dynamics of a second order differential equation and estimate uncertainty at the same time. We do this by creating a custom loss function that trains the model to approximate the second derivative of a noisy time series, and to penalize based on a separate parameter which corresponds to the desired quantile. The resulting gradient boosting model simulates stochastic trajectories of the system, given a single starting point; that is, it estimates the expected trajectory and its uncertainty. We show that the uncertainty estimation is well calibrated, and that the model can learn the dynamics even in the presence of noise. The approach is demonstrated on a simple cartpole system.

II. Problem Setting

In developing and testing a UDE capable of uncertainty quantification, we use the Cartpole model, for the following reasons:

- A known model allows us to simulate data and add noise while knowing ground truth. While the knowledge of the ground truth is a strong assumption in realistic applications, this aspect was important in a proof of concept.
- The differential equations for the cartpole model are known but nonlinear and complex. Therefore, we hypothesized that approximating the equations to a high degree of fidelity would be a difficult task.

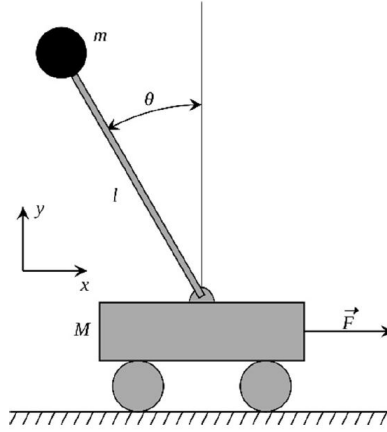


Fig. 1 Test case: Cartpole with pendulum.

We use the following process:

- 1) Use the cartpole differential equations to generate trajectories for θ , the angle of the pendulum attached to the cart; x , the horizontal distance of the cartpole from the starting point; and the first derivatives of both.
- 2) Use the values of θ and x at every time step as training features, and the difference between the current value and the value at the next time step as the target.
- 3) Train the model on these data.
- 4) Given a new starting condition, predict a full conditional distribution for the change in each variable over the next time step. Sample from this distribution and repeat this process many times to generate a single predicted trajectory.
- 5) Repeat step 4 many times to generate many trajectories, which together define a predicted distribution for all variables over time, given specified starting conditions.

We tested dynamic quantile regression with gradient-boosted trees using this setting. The model is explained in more detail in section VI.

III. Cartpole model

The cartpole model used here is based on Cannon et al. [3]:

$$\ddot{x} = \frac{f - ml\ddot{\theta} \cos\theta + ml\dot{\theta}^2 \sin\theta + F_f}{M}$$

$$\ddot{\theta} = \frac{Mg \sin\theta - \cos\theta (f + ml\dot{\theta}^2 \sin\theta + F_f) - \frac{MM_f}{ml}}{ml \cos^2\theta - (1+k)Ml}$$

where

m = Mass of the pole (kg)

m_c = Mass of the cart (kg)

M = Combined mass of the pole and the cart (i.e., $M = m + m_c$)

l = Length of the pole (m)

g = Constant of gravitation (m/s^2)

f = Input force applied to the cart to push it to the left or right (N). A positive value means a push to the right and a negative value means a push to the left.

x = Horizontal position of the cart on the track as measured from the origin (m)

$\dot{x}, \ddot{x}$ = Velocity and acceleration of the cart, respectively

θ = Angle of the pole (rad)

$\dot{\theta}, \ddot{\theta}$ = Angular velocity and acceleration of the pole, respectively (rad/s)

The rotational inertia of a point mass (m) rotating at a fixed distance l from the axis of rotation is $J = ml^2$. Cannon et al. [3] use $J = ml^2/3$, which is the moment of inertia for a rod whose mass is uniformly distributed along its length. Other definitions of J are multiples of ml^2 ; hence, k is used to express the moment of inertia in a general form and an appropriate value of k should be substituted in simulation. In the present case, $\frac{1}{3}$ was used in place of k during simulation, following Cannon's convention.

The two friction terms, F_f and M_f , appearing in the above equations, indicate the cart-track and cartpole frictions, respectively. Following Cannon's convention, F_f is considered proportional to the velocity $\dot{x}$ of the cart (i.e., $F_f = -\mu_c \dot{x}$) and M_f to the angular velocity ($\dot{\theta}$) of the pole (i.e., $M_f = \mu_p \dot{\theta}$). It is worth noting that many complex friction models can account for the cart-track and the cartpole friction in the dynamic model; however, we use simpler ones to avoid any discontinuities.

IV. Data Collection

The dynamic model described in section III was used to generate the dataset for training the gradient-boosting model. To run the cartpole model, the following physical parameters were chosen arbitrarily: $m = 0.1$ kg, $m_c = 1$ kg, $l = 0.5$ m, $k = \frac{1}{3}$, $\mu_c = 0.0005$, and $\mu_p = 0.000002$.

For this paper, we used three trajectories: training, validation, and test. The initial conditions for all three are shown in Table IV. The validation trajectory is an extension of the training trajectory, while the test trajectory is completely new and has a different control policy as well. The control policies were chosen to facilitate thorough exploration of the feasible parameter space.

Variable	Train	Validation	Test
θ	0.3	209.4	-0.1
$\frac{d\theta}{dt}$	1.0	-3.4	-3.0
x	0.0	-41.0	0.0
$\frac{dx}{dt}$	-0.72	0.37	0.15
Control Policy	$F(t) = 5 \sin(6t)$	$F(t) = 5 \sin(6t)$	$F(t) = -4 \cos(4.5t)$
Seconds	500	100	100

We used the Runge-Kutta 45 scheme to solve the initial value problem (IVP) for each trajectory, saving the data at a time step of 0.01 seconds. Our initial goal was to evaluate at all 100 seconds of the validation and test data. However, we

changed evaluation to the first 10 seconds, since forecasts from the gradient-boosted model had prediction intervals that diverged before that time. Finally, we generate the data three times: once with no noise, once with 0.1 scale Gaussian noise, and once with 0.5 scale Gaussian noise.

V. Evaluation

An important component of our process is the quantitative determination of the method’s predictive accuracy. We are more concerned with distributional accuracy than point accuracy. We chose to use the interval score to evaluate various predictions, also referred to as the Winkler score.[4] It is used to evaluate the lower and upper bounds of a single prediction interval. In brief, there are two components: (1) the width of the interval (smaller is better), and (2) the distance of an observation from the interval. If the observation is within the interval, the second component is zero. When the observation is outside the interval, the distance from the interval is weighted more than the width of the interval by a factor of $\frac{2}{\alpha}$, where the prediction interval’s lower and upper bounds are $\frac{\alpha}{2}$ and $1 - \frac{\alpha}{2}$, respectively. Equation 1 shows the full calculation, where p_u and p_l are the upper and lower prediction intervals, y is the observation, and 1 is the indicator function.

$$Score = (p_u - p_l) + \frac{2}{\alpha}(p_l - y)1(y < p_l) + \frac{2}{\alpha}(y - p_u)1(y > p_u) \quad (1)$$

We evaluate the 50%, 80%, and 95% prediction intervals on the first 10 seconds of each trajectory, computing and averaging the interval score at every time step for all three intervals.

VI. Methods

The LightGBM[5] library is used to implement quantile regression with gradient-boosted trees to approximate differential equations. To the authors’ knowledge, the approach has not been used in this domain. The proposed approach enables estimation of the full conditional distribution, which allows for robust uncertainty quantification. The subsections, below, describe the component algorithms in more detail.

A. Gradient Boosting

Gradient boosting is a machine learning technique typically used in supervised learning tasks. It computes predictions from an ensemble of many weak learners. When the weak learners are decision trees, the algorithm is known as gradient-boosted trees. It usually outperforms other tree-ensemble methods, such as Random Forest, and is among the most common algorithms used in machine learning competitions. It operates by repeatedly building decision trees to optimize a given cost function. The algorithm has been described as functional gradient descent, or gradient descent in function space. While gradient boosting lacks some of beneficial properties, such as differentiability, modern gradient-boosting libraries have the ability to learn patterns from tabular data faster and more reliably than any other modeling framework. The ability to provide a custom loss function also enables these models to work in a wide variety of applications, including approximating differential equations.

B. Dynamic Quantile Regression

Quantile regression is a sub-field of regression that aims to predict quantiles of a conditional response variable, such as the median, as opposed to the conditional mean. Typically, a single quantile regression model is trained to predict a single quantile, by using the loss function $\rho_\tau(m) = m(\tau - 1_{m < 0})$, where 1 is the indicator function. A recent advance in quantile estimation techniques includes the desired quantile as a feature in the training data, as well as a custom loss function that calculates the row-wise gradient based on the desired quantile in each row, e.g., [6]. We call this approach dynamic quantile regression. Gradient-boosted tree frameworks, moreover, allow for imposing monotonicity, which can be used to ensure non-crossing quantiles. Gradient-boosted trees using this method of quantile regression can estimate the entire conditional distribution at once, provided that a sufficient number of quantiles are supplied in training.

C. Additional Attributes

In the presence of noise, we found the algorithm to perform better with smoothing applied. We used a simple moving average as a smoothing technique. We used two additional pieces of information to ease learning the dynamics:

we removed the position of the cart from the model's inputs, since it obviously has no effect; and we forced θ to be in the interval $[0, 2\pi]$. We test and show the results with and without the second change.

VII. Computational Results

Overall, gradient-boosted trees are able to learn the dynamics of the cartpole system to a degree. When using the model to forecast the dynamics by solving IVPs, which we call multi-step predictions, the error propagation leads to increasing uncertainty after a few hundred time steps. However, the uncertainty bands are reasonably well-calibrated even when the signal is lost.

A. Multi-Step Predictions

Model results for the gradient-boosted trees are shown in Fig. 2 below. Under all three starting conditions, the predictions are very accurate and confident in the first couple of seconds, which corresponds to the first couple hundred data points. Prediction intervals start to diverge from 2-5 seconds after initialization, but importantly still capture the uncertainty well.

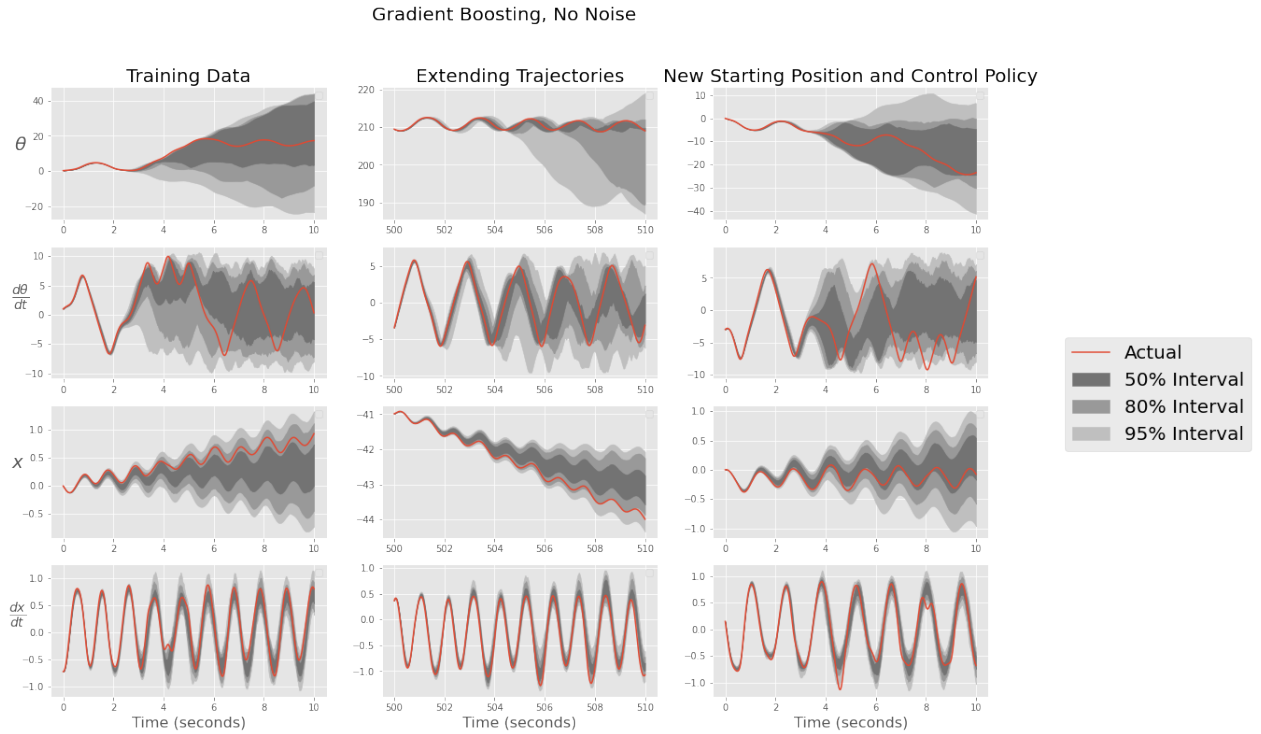


Fig. 2 Multi-step predictions for the Gradient Boosting Model compared to actuals. Only the starting position is given to the model.

Fig. 3 shows the interval score as a function of time for each variable on the three multi-step tasks. The performance gets worse as a function of time, as expected, but generally continues to be reasonably stable.

B. Differential Equation Approximation

The gradient-boosted trees show close correspondence to the true differential equation, depending on the level of noise and amount of information provided to the model. Fig. 4 shows the partial dependency between θ and $\frac{d^2\theta}{dt^2}$. We focus on this comparison because θ has the largest influence on $\frac{d^2\theta}{dt^2}$ under our experimental conditions. The majority of the predicted distribution is centered on the correct values, although the 95% prediction interval does extend to values of much smaller magnitudes when $\frac{d^2\theta}{dt^2}$ is high. Predictions are irrelevant outside the interval $[0, 2\pi]$, since we converted θ to this range prior to running model predictions.

Interval Scores, Gradient Boosting, No Noise

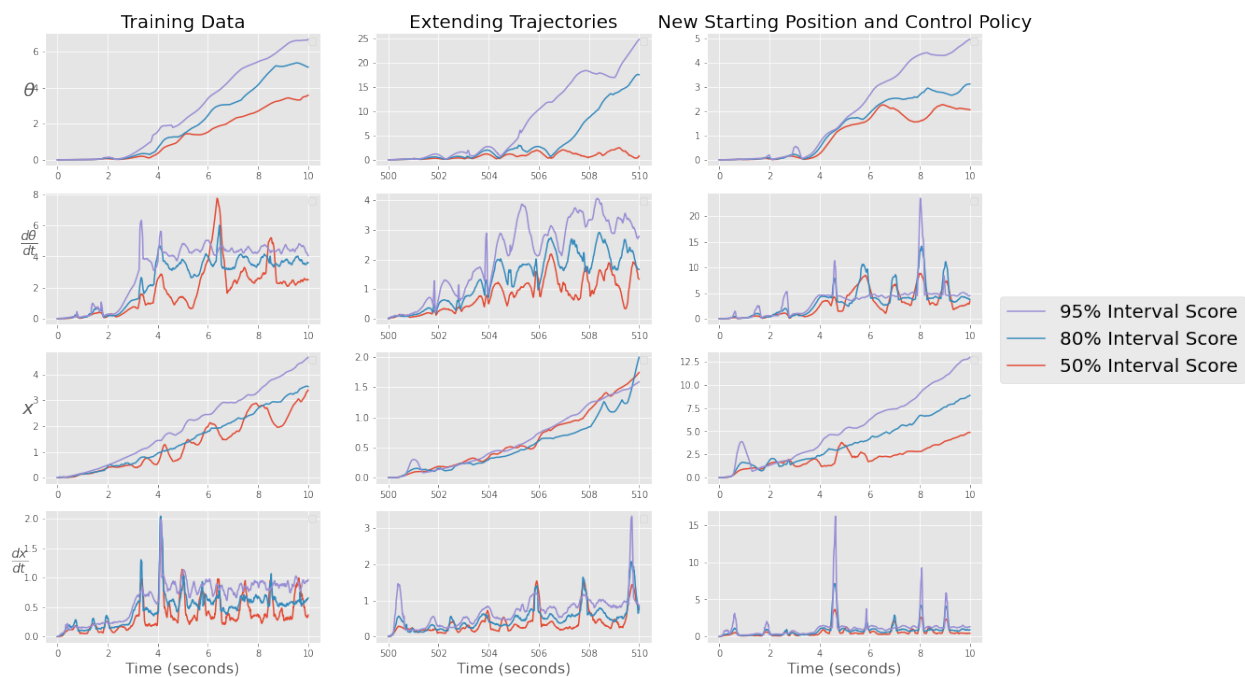


Fig. 3 Interval score for multi-step predictions for the Gradient Boosting Model.

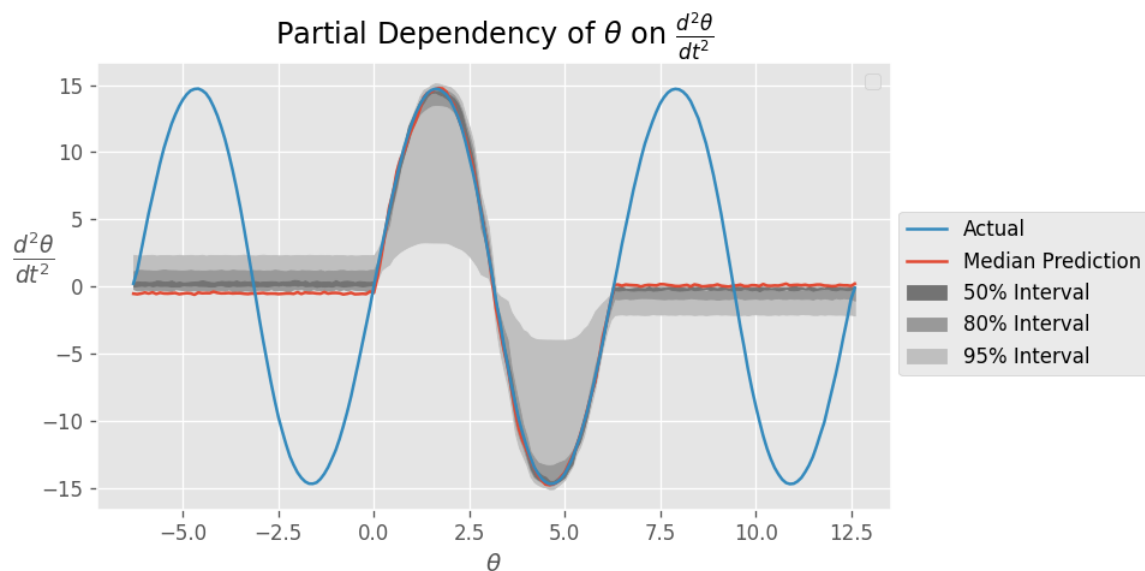


Fig. 4 Partial dependency plot for θ on $\frac{d^2\theta}{dt^2}$ from the Gradient Boosting Model compared to the true differential equation. θ is converted to the interval $[0, 2\pi]$ before being passed into the model. This is the model trained on the data with no noise.

C. Approximation Performance of Differential Evolution across Various Model Parameters

We noticed differences in how well the gradient boosting model approximated the true DE given what model parameters were used. We briefly cover some of those in this section.

Converting θ to the interval $[0, 2\pi]$

Most importantly, when θ was not converted to the interval $[0, 2\pi]$, the model was not able to learn the DE with as much precision, but instead had to re-learn the same relationships in every interval $[k\pi, (k+2)\pi]$. This phenomenon is seen clearly in Fig. 5, where the top row corresponds to our chosen model, and the bottom to the model that doesn't first convert θ . The bottom row shows the wider prediction intervals at especially the 50% and 80% levels, which is exacerbated when the noise is increased.

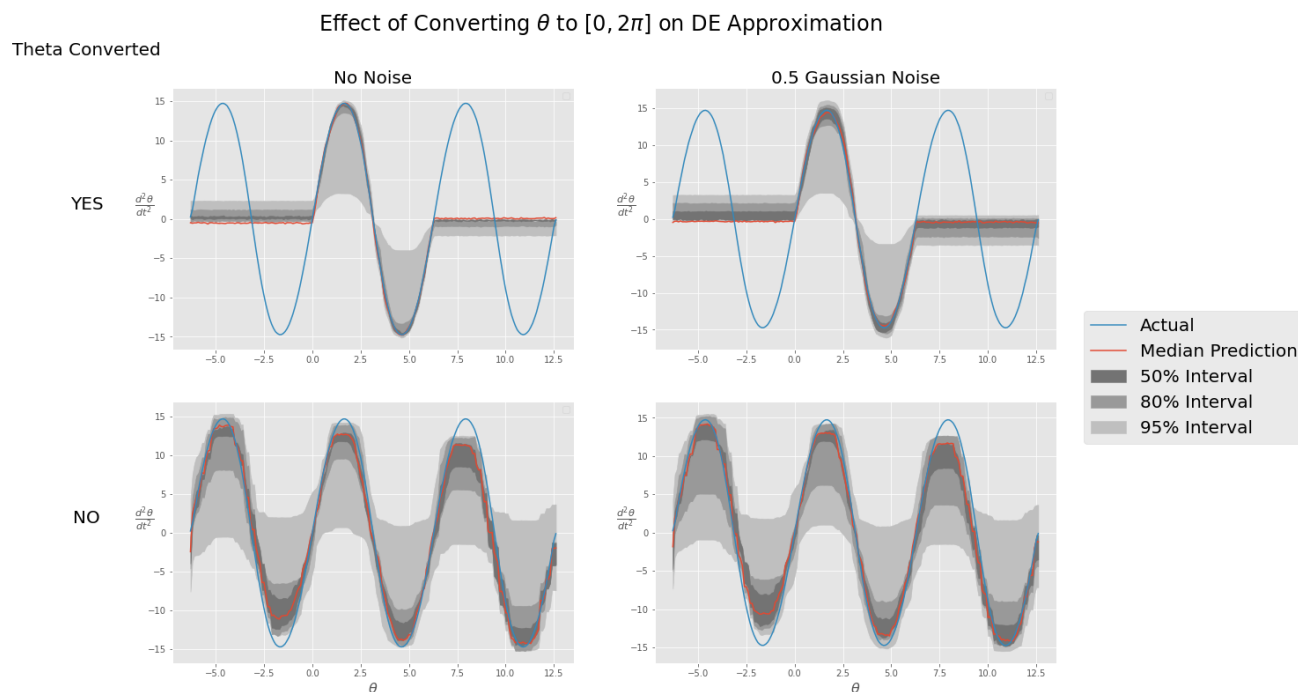


Fig. 5 Partial dependency of θ on $\frac{d^2\theta}{dt^2}$ with and without converting θ to the interval $[0, 2\pi]$. Left and right columns are no noise and 0.5 scale gaussian noise, respectively.

Effect of Training Iterations and Data Size

Fig. 6 compares the DE approximations from gradient boosting models with different numbers of training iterations or rounds, and seconds of training data provided. The differences are small, but can be most clearly seen in the proximity of the lower 95% interval to the true equation vs 0. Interestingly, the number of training rounds seems to matter more than the amount of training data provided to the model.

Effect of Data Smoothing

We found simple mean-smoothing to improve performance of the gradient boosting model when training on noisy data. Fig. 7 compares the performance of no smoothing to windows of 5 and 15 for mean smoothing. We have 100 time steps per second. Thus, these window sizes correspond to 0.05 and 0.15 seconds, respectively. The model has significantly less precision on noisy data when no smoothing is performed. The differences between the two window sizes are more subtle. With the high window size, the prediction distribution is slightly lower in magnitude when $\frac{d^2\theta}{dt^2}$ is at a local minimum or maximum. This happens because the data are smoothed across the local minimum or maximum, making it consistently slightly smaller than it truly is. For these reasons, we chose to use a window size of 5.



Fig. 6 DE approximation compared across number of training rounds and training seconds. Differences most clearly seen in lower 95% interval proximity to the true equation.

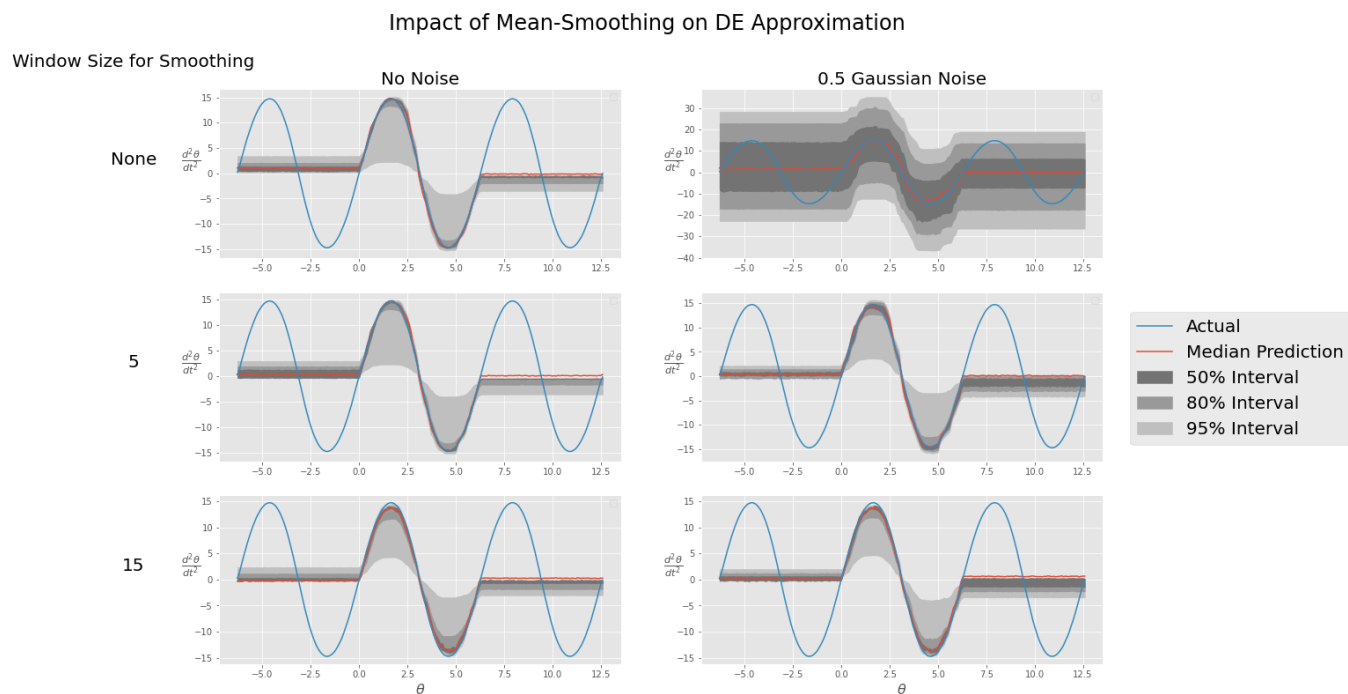


Fig. 7 DE approximation compared across smoothing window size and with or without noise in the training data.

VIII. Concluding Remarks

In this work, we demonstrated the ability of dynamic quantile regression with gradient-boosted trees to produce accurate predictions and uncertainty quantification for the cartpole system when used to approximate the full differential equations. We are cautiously optimistic about this result because in many application domains, the governing differential equations are at least partially known, yet operate under the conditions of noise that is difficult or impossible to account for. Gradient-boosting can be used in these domains to incorporate both the known physics around a system and real data from that system. While the cartpole system is a simple problem, we tasked the proposed method with fully approximating the differential equations. The performance of the approach during this preliminary proof of concept suggests that the method is likely to perform well in more complicated systems if we provide more information to the model about the known dynamics of the system. The next steps are the study of the methodology on increasingly complex problems, as well as the representation of the uncertainty to human team members in CPH teams.

Acknowledgments

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